Chapter 7-8 Principal Component Analysis and Factor Analysis

1(12 points). Let the correlation matrix of $\mathbf{x} = (x_1, x_2, \cdots, x_p)'$ be

$$R = \begin{bmatrix} 1 & 0.5 & \cdots & 0.5 \\ 0.5 & 1 & \cdots & 0.5 \\ \vdots & \vdots & \ddots & \vdots \\ 0.5 & 0.5 & \cdots & 1 \end{bmatrix}.$$

Such a correlation matrix is commonly used to describe relationships among ecological variables, such as biological size measurements. Determine the eigenvalues of R (3 points), the orthonormal unit eigenvectors (6 points), and the contribution rates of the principal components (3 points).

2. Please prove that the sum of elements square of $\boldsymbol{S} - (\hat{\boldsymbol{A}}\hat{\boldsymbol{A}}' + \hat{\boldsymbol{D}})$ is less than $\hat{\lambda}_{m+1}^2 + \cdots \hat{\lambda}_p^2$ in equation (8.3.2).

[Hints: the sum of squares of elements in $S - (\hat{A}\hat{A}' + \hat{D})$ is less than or equal to the sum of squares of elements in $S - \hat{A}\hat{A}'$]

3. 有中国各省区八项男子竞赛运动记录数据，指标分别用$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ 表示，通过 SAS 统计软件分析得到的部分结果如下所列。

The SAS System

The FACTOR Procedure

Initial Factor Method: Principal Components

Prior Communality Estimates: ONE

Eigenvalues of the Correlation Matrix: Total= 8 Average = 1				
	Eigenvalue	Difference	Proportion	Cumulative
1	6.62214613	5.74452784	0.8278	0.8278
2	0.87761829	0.71829715	0.1097	0.9375
3	0.15932114	0.03527176	0.0199	0.9574
4	0.12404939	0.04416911	0.0155	0.9729
5	0.07988027	0.01191512	0.0100	0.9829
6	0.06796515	0.02154562	0.0085	0.9914
7	0.04641953	0.02381943	0.0058	0.9972
8	0.02260010		0.0028	1.0000

The FACTOR Procedure

Rotation Method: Varimax

Orthogonal Transformation Matrix		
	1	2
1	0.75888	0.65124
2	-0.65124	0.75888

Rotated Factor Pattern		
	Factor1	Factor2
x1	0.27430	0.93519
x2	0.37644	0.89291
x3	0.54306	0.77252
x4	0.71240	0.62670
x5	0.81333	0.52539
x6	0.90194	0.38881
x7	0.90346	0.39651
x8	0.93554	0.26095

Scoring Coefficients Estimated by Regression

Squared Multiple Correlations of the Variables with Each Factor	
Factor1	Factor2
1.0000000	1.0000000

Standardized Scoring Coefficients		
	Factor1	Factor2
x1	-0.30042	0.53957
x2	-0.22153	0.45922
x3	-0.06771	0.29112
x4	0.10008	0.10337
x5	0.20712	-0.01890
x6	0.32436	-0.16055
x7	0.32148	-0.15576
x8	0.40598	-0.26906

(1) 求两个公因子（factor1 和 factor2）解释的方差大小；

(2) 请写出旋转后的因子载荷矩阵，并计算各个变量的共性方差的大小；

(3) 因子载荷矩阵里第二行元素的平方和（记为 h_2^2 ）所表达的含义；

(4) 作因子分析时因子旋转的目的是什么？请写出上述因子旋转时的正交变换矩阵。

(5) 常用的因子得分的估计方法有哪种？根据题中所列结果，请写出标准化后的因子得分计算公式。